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Effects of cross baffles on the flow of the microalgae in raceway photobioreactors

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ABSTRACT

This paper focuses on the increase in the mixing degree of the fluid in the raceway photobioreactors and puts forward a new type of cross baffles. Computational fluid dynamics is adopted to study the fluid mechanics and the trajectory of the microalgae in the raceway photobioreactors. The research result indicates that cross baffles, compared with other baffles, bring better distribution of the vortex flow, higher vorticity, and a more favorable proportion of the dead zone. In addition, it is found that in the raceway photobioreactors with cross baffles, the light-dark cycle is reduced by 18%, 5.9%, and 7.3% respectively than that of the raceway photobioreactors with other three types of baffle plates. Meanwhile, the outdoor cultivation experiment of *Chlorella* proves that the optimized cross baffle which can enhance the mixing effects is able to increase the biomass concentration of the RWP by 11.1%.

1. Introduction

To mitigate the global warming and greenhouse effect caused by excess emissions of carbon dioxide, fossil fuels are gradually being replaced by renewable energy sources. Microalgae, which boast great potential in carbon dioxide sequestration [1], bio-feed [2], recycling of heavy metals [3], and the pharmaceutical and nutraceutical industry [4], are coming into the spotlight. The photobioreactors are one of the core devices of microalgae culture system, and the raceway photobioreactors(RWP), developed based on the concept of "high rate algal ponds (HRAP)" by Oswald featuring simple structure [5], low cost and easy operation, are the main photobioreactors used in current large-scale commercial production [6]. Therefore, how to enhance the mixing degree of the fluid in the RWP is a major focus for researchers.

To increase the mixing degree of the fluid in the RWP, researchers conducted studies on the structure of the RWP. According to their studies, adding internals to the RWP is a common way to promote the mixing. The comparative study by Kusmayadi [7] found that the addition of CO_2 sprayers can effectively reduce the low velocity zone within the RWP and increase the production of microalgae, but has no remarkable effect in the mixing of the fluid in the RWP. Moreover, the low liquid level leads to an extremely low carbon dioxide utilization by microalgae. Therefore, it is not suitable for large-scale production of microalgae. Cheng [8] put up with a combination of upward baffle plates and downward baffle plates. The fluid can be mixed to a greater extent by flowing through the upward baffle plate and the downward baffle plate successively to generate clockwise and counterclockwise vortex flows. However, such a structure will result in a low degree of mixing and local aggregation of algal cells in the backflow area of the baffle, unfavorable for the mixing of cells there. Ali Kubar [9] devised another kind of baffle plates, namely the helical baffle. It changes the fluid from a direct current to a helical flow through the twisting rotation of the baffles while forming vortex flows around the baffle to optimize the mass transfer efficiency of the algal solution, but the space-consuming complicated structure make it an undesirable choice.

There are another two types of RWPs. In order to increase the mixing degree of the fluid in the reactor and decrease the energy consumption, Huang [10] designed a new airlift-driven sloping raceway pond which combines the airlifting system and the sloping raceway pond. While the mixing degree proved to be improved with the new structure, Huang

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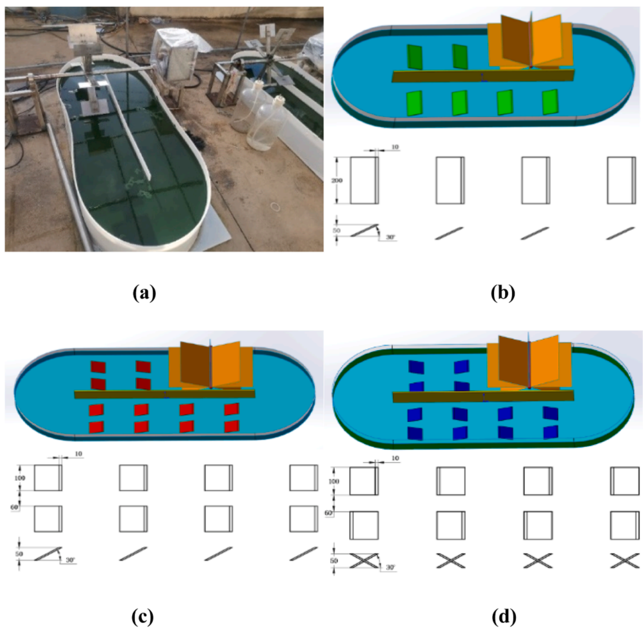


Fig. 1. (a) RWP outdoor model (b)Structure B (c) Structure C (d) Structure D.

concluded less energy was consumed without detailed analysis. In spite of that, the new structure is large in the overall size. Therefore, it is not applicable to large-scale outdoor cultivation. The other type was proposed by Huang [11]. They came up with a V-shaped bottom to increase the conductivity and the ratio of surface area to volume. By adjusting the inclination angle of the V-shaped groove, the depth of the algae solution, and the air intake speed, Huang Wen and his team compared the motion of the fluid, and found the change to the RWP led to little increase in the mixing performance and absence of local vortex flows.

The computational fluid dynamics (CFD) based numerical simulation is conducive to the design and optimization of the fluid dynamics in the RWP. Sawant [12] figured out, with the help of CFD, the effect of the length-to-width ratio of the channel and the position of the axial flow impeller for side entry on the fluid dynamics in the RWP. Belohlav [13] adopted CFD when designing his research in the fluid dynamics in the mixed horizontal tubular photobioreactors used for microalgae culture and wastewater treatment. He evaluated the flow regime, the mean cycle time, and the distribution of shear stress, and verified the CFD model via tracer experiment and velocity measurement by ultrasonic flowmeter. Olmo [14] coupled CFD with dynamic photosynthesis model to analyze the photosynthetic reaction efficiency in the ring-shaped photobioreactors of different sizes and circulation rates.

Although there are many studies on the mixing performance of the RWP, how to improve the mass transfer efficiency of microalgae remains a major subject of research. This paper proposed a novel structure of cross baffles to advance the mixing of the fluid and speed up the growth of microalgae. CFD and comparative study were applied to analyze the distribution of the vortex flow, the vorticity, the trajectory of the fluid, and the mixing effect of the inner flow field in the RWPs with different baffles, as well as the fluid dynamics and light regime in the RWP with cross baffles. Based on the analysis, the structure of the cross baffle was optimized to reinforce the mixing effect of the inner flow field of the RWP. Then an outdoor cultivation experiment was conducted to verify the optimized structure.

2. Materials and methods

2.1. 3D model and meshing

Fig. 1 shows the outdoor model of the RWP. The RWP was made of a 6 mm thick polyethylene sheet, with electric paddlewheels made of steel plates and baffles of plexiglass. RWP without baffle is structure A. Fig. 1 also displays the three structures of baffles, including the cross baffles (Fig. 1(d)) developed to increase the mixing effect and the mass transfer capability of the RWP. ICEM was employed for tetrahedral mesh. In order to avoid the impact of the mesh size and the mesh number on the results, three different sizes of meshing (621,258, 1,012,386, 1,215,684) were provided for RWP without baffles. The results showed that under the working condition of speed of paddlewheel of 20 rpm, there was only 0.78% deviation between the calculated values of 1,215,684 and 1,012,386 mesh numbers. Considering this, a model with the mesh number of 1,012,386 was selected for calculation.

The numerical simulation in this paper was achieved by FLUENT. The pressure solver was used for the solution of unsteady state. With a time step length of 0.05 s and a total of 12,000 steps, when the residual curve was under 10^{-6} and the selected speed monitoring parameters changed periodically within a certain range, convergence was determined to have been reached. VOF (volume of fluid) implicit model was used to simulate the gas-liquid two-phase flow, in which water was the main phase and air the second. The VOF model was configured as below: the outlet of the RWP was set as the pressure outlet, the blade and the stirring shaft as Moving wall, and the other parts as wall boundary conditions; the body force weighted scheme was activated to enhance the stability of the calculation considering the impact of gravity; a k-ε model was chosen for the turbulence model; and enhanced wall treatment was selected. Sliding mesh was adopted to simulate the dynamic and static zones surrounding the stirring paddlewheel. The ambient temperature was set to 25 °C, and the influence of heat exchange was ignored.

The governing equation for this study is as follows:

Continuity equation

$$\frac{\partial\rho}{\partial t}+\frac{\partial(\partial\rho u_x)}{\partial x}+\frac{\partial\left(\partial\rho u_y\right)}{\partial y}+\frac{\partial(\partial\rho u_z)}{\partial z}=0 \tag{1}$$

In this formula, t(s) is the fluid motion time, ρ (g/m3) is the fluid density, $\boldsymbol{u}_x$, $\boldsymbol{u}_y$, and $\boldsymbol{u}_z$ (m/s) represent the axial velocity components of x, y and z.

Momentum quation

$$\frac{\partial(\rho u_x)}{\partial t}+\nabla\cdot(\rho u_x\vec{u})=-\frac{\partial p}{\partial x}+\frac{\partial\tau_{xI}}{\partial x}+\frac{\partial\tau_{yx}}{\partial y}+\frac{\partial\tau_{zx}}{\partial z}+\rho f_x \tag{2}$$

$$\frac{\partial\left(\rho u_y\right)}{\partial t}+\nabla\cdot\left(\rho u_y\vec{u}\right)=-\frac{\partial p}{\partial y}+\frac{\partial\tau_{xy}}{\partial x}+\frac{\partial\tau_{yy}}{\partial y}+\frac{\partial\tau_{zy}}{\partial z}+\rho f_y \tag{3}$$

$$\frac{\partial(\rho u_z)}{\partial t}+\nabla\cdot(\rho u_z\vec{u})=-\frac{\partial p}{\partial x}+\frac{\partial\tau_{xz}}{\partial x}+\frac{\partial\tau_{yz}}{\partial y}+\frac{\partial\tau_{zz}}{\partial z}+\rho f_z \tag{4}$$

Here, p (Pa) refers to the pressure of the fluid in the control body; $\boldsymbol{u}$(m/s) is the fluid velocity;$\boldsymbol{\tau}_{xx}\boldsymbol{\tau}_{xy}$and $\boldsymbol{\tau}_{xz}$are the component of viscous stress on the surface; f_x, f_y and f_z(N) are the volume forces on the control body.

2.2. Vortex structure analysis

The vortex flow in the flow field has prominent impact on the flow of the microalgae in the photobioreactors. According to Euler's method, a common way to identify the vortex flow is the Q-criterion proposed by Hunt [15], which points out that the deformation of fluid can be represented by the velocity gradient tensor$\frac{\partial u_i}{\partial x_j}$. The tensor can be decomposed into two parts:

Symmetric strain rate tensor

$$S_{ij}=\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right) \tag{5}$$

Antisymmetric vortex tensor

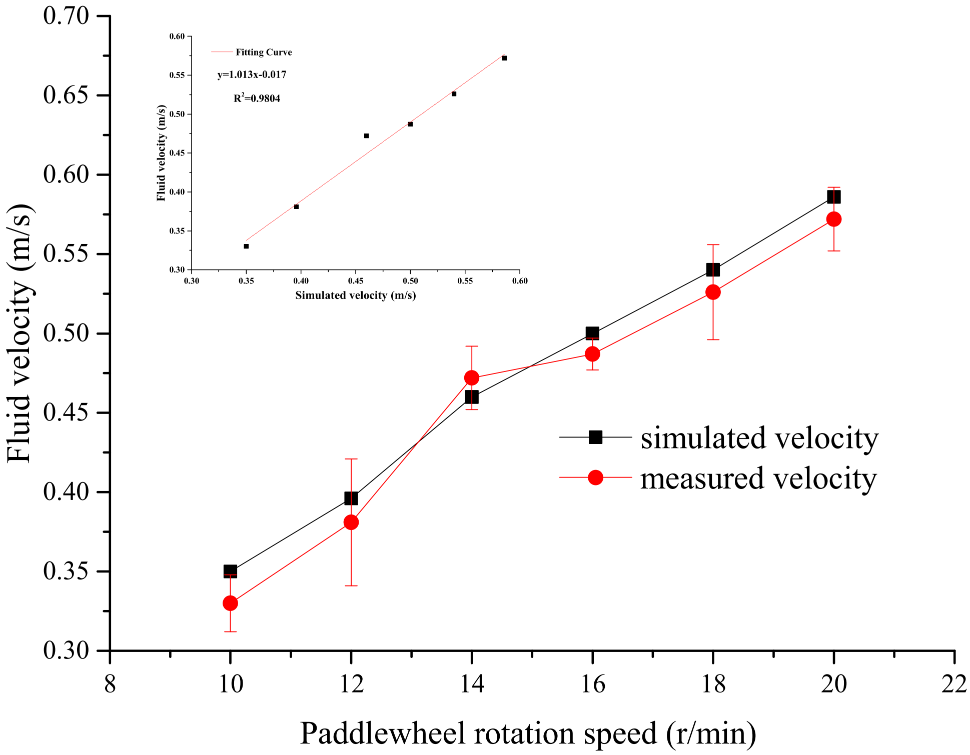


Fig. 2. Fitting CFD simulated flow velocity with measured flow velocity.

$$\Omega_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right) \tag{6}$$

If the influence of the vortex tensor on the deformation of the fluid outweighs that of the strain rate tensor, it is determined that the vortex flow exists in that area. Thus,

$$Q = \frac{1}{2}\left(\| \Omega\|^2 - \| S\|^2\right) = -\frac{1}{2}\frac{\partial u_i}{\partial x_j}\frac{\partial u_j}{\partial x_i} \tag{7}$$

In the above equation, $S$is the symmetric strain rate tensor; $\Omega$is the antisymmetric vortex tensor; $u_i$and $u_i$are the velocity components in i and$j$ directions;$\| \ \|$ is the 2-Norm of the tensor, and Q, whose size is independent of the choice of coordinate system, is the criterion in s^{-2}. When $Q > 0$ in the flow field, the vortex tensor contributes more to the deformation of the fluid in RWP than the strain rate tensor does, which is identified as the vortex. However, Q-criterion, even boasting the advantage of Galilean invariance, cannot identify the vortex structure correctly in the rotation and acceleration reference frame. Nevertheless, Q-criterion is applicable here for there is no rotation or acceleration in the direct current channel in a RWP model with the open cavity flow.

2.3. Force analysis of particles

This paper adopts dynamical proton model (DPM) of ANSYS FLUENT to describe the motion of microalgal cells in RWP as the movement of particles in a fluid. By solving the differential equation for the forces acting upon particles in a fluid under the Lagrange coordinate system using the integrating factor method, DPM is applied to track the trajectory of the particles. The force equilibrium equation for the particles is as below [16]:

$$\frac{du_p}{dt} = F_D\left(\vec{u} - \vec{u}_p\right) + \frac{\vec{g}\left(\rho_p - \rho\right)}{\rho_p} + \vec{F} \tag{8}$$

$$F_D = \frac{18\mu}{\rho_p d_p^2}\frac{C_D R_e}{24} \tag{9}$$

where $\vec{u}$(m/s) is the fluid velocity, $\vec{u}_p$(m/s) is the particle velocity, μ(N•s/m^2)为is the dynamic viscosity of the fluid, ρ(kg/m^3) is the density of the fluid, ρ_p(kg/m^3) is the density of the tracked particles, d_p(m) is the particle diameter,$R_e$is the Reynolds number, calculated as $R_e \equiv \frac{\rho d_p |u_p - u|}{\mu}$. $C_D$is the drag coefficient. $F_D(\vec{u} - \vec{u}_p)$(N) is the drag force on the particle per unit mass, $\frac{\vec{g}(\rho_p - \rho)}{\rho_p}$ (N) the gravity, and $\vec{F}$ (N) the additional forces such as lift force and virtual mass force. Since the drag and buoyancy force on *Chlorella* cells in the algal solution are much greater than the lift force and virtual mass force, the additional forces are ignored, and only the resistance and the pressure gradient force are considered [17].

The equation for the trajectory of the particles is:

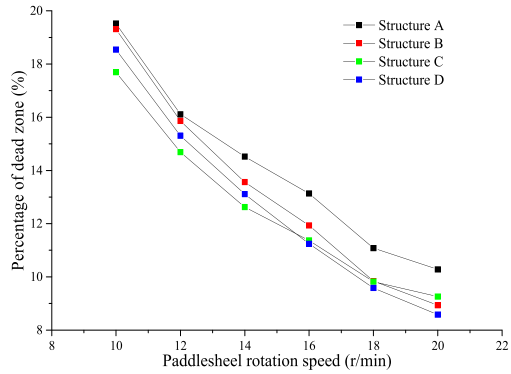
$$\frac{dx}{dt} = u_p \tag{10}$$

In the above equation,$x$is the particle motion distance; $t$is the particle motion time, and u_p is the particle motion speed. The trajectory of the particles is depicted by their position coordinates at different times which are obtained by solving the equation based on directions and the size of discrete time step.

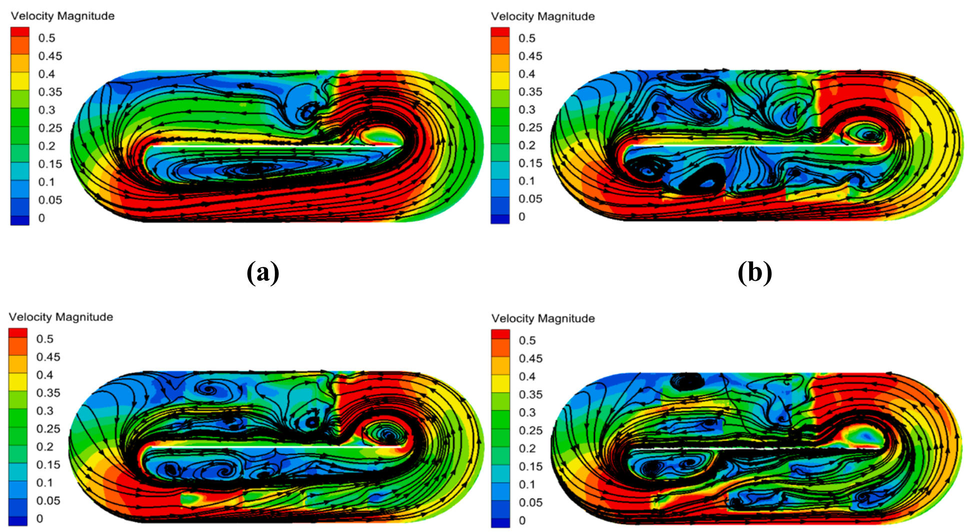
2.4. Mixing time and mass transfer coefficient

The mixing time was measured by monitoring the change in the conductivity of the RWP. Water was added to the RWP and the conductivity probe was put in the direct current channel. As the initial conductivity became stable, sodium chloride solution was added to the front of the paddlewheel. After the liquid was mixed evenly, the conductivity of the solution turned stable and reached a value 5% lower than the final value.

The calculation formula for mass transfer coefficient can be found in reference [18]. The following method was employed to measure the volumetric mass transfer coefficient. Nitrogen was injected into the RWP to reduce the dissolved oxygen concentration in the water to 2.5 mg/L, then the aerator was used to inject air into the water and the web test



(a) Ratio of dead zone of four structures



(a) **(b)**

(c) **(d)**

(b) Streamline distribution of four structures

Fig. 3. (a) Ratio of dead zone of four structures. (b) Streamline distribution of four structures.

system recorded the rising dissolved oxygen concentration in the liquid in the RWP. The mass transfer coefficient was calculated according to the rising dissolved oxygen concentration.

3. Result and discussion

3.1. Simulation verification

To verify the accuracy of the numerical simulation, the flow velocity of Structure A was measured by the current meter and compared with the calculated flow velocity using CFD. Fig. 2 displays the fitting curve of the calculated flow velocity and the measured flow velocity. As it shows, the calculated flow velocity is linearly correlated with the measured flow velocity, and the fitting degree is about 0.9804, indicating that the CFD simulation is capable of reflecting the motion in the inner flow field of the RWP.

3.2. Velocity analysis of flow field in RWP

The velocity dead zone here is defined as the region in the reactor where the flow velocity is lower than 0.1 m/s [19]. The formula is:

$$x = \frac{V_{v<0.1}}{V_{pond}} \times 100\% \tag{11}$$

$V_{v<0.1}$is the volume of the area where the flow rate of water is less than 0.1 m/s, V_{pond} is the total volume of water in RWP, $x$is the proportion of dead zone.

Algal cells will deposit in the dead zone, with rising dissolved oxygen concentration affecting their normal metabolism. Therefore, the dead zone must be taken into consideration when designing baffles. Fig. 3(a) presents the proportion of velocity dead zone in the RWPs with different structures at different rotation speeds of the paddlewheel. As the speed increases, the proportion of the velocity dead zone in the RWP decreases. When the rotation speed of the paddlewheel reaches 20 r/min, the RWP with the cross baffles has the least proportion of the velocity dead zone, 16.5%, 4%, and 8% smaller than that of the RWPs, respectively, without baffles, with baffles of Structure B, and with baffles of Structure C.

Fig. 3(b) represents the velocity distribution and the streamline at Z = 35 mm in the RWPs with different structures. As it shows, the centrifugal force creates a pressure difference at the curve, leading to high-velocity zones in the outer side of the baffle and a low-velocity zone near the baffle, and Structure B and Structure C are unable to prevent such a phenomenon from presence. Instead, in the RWPs with Structure B and

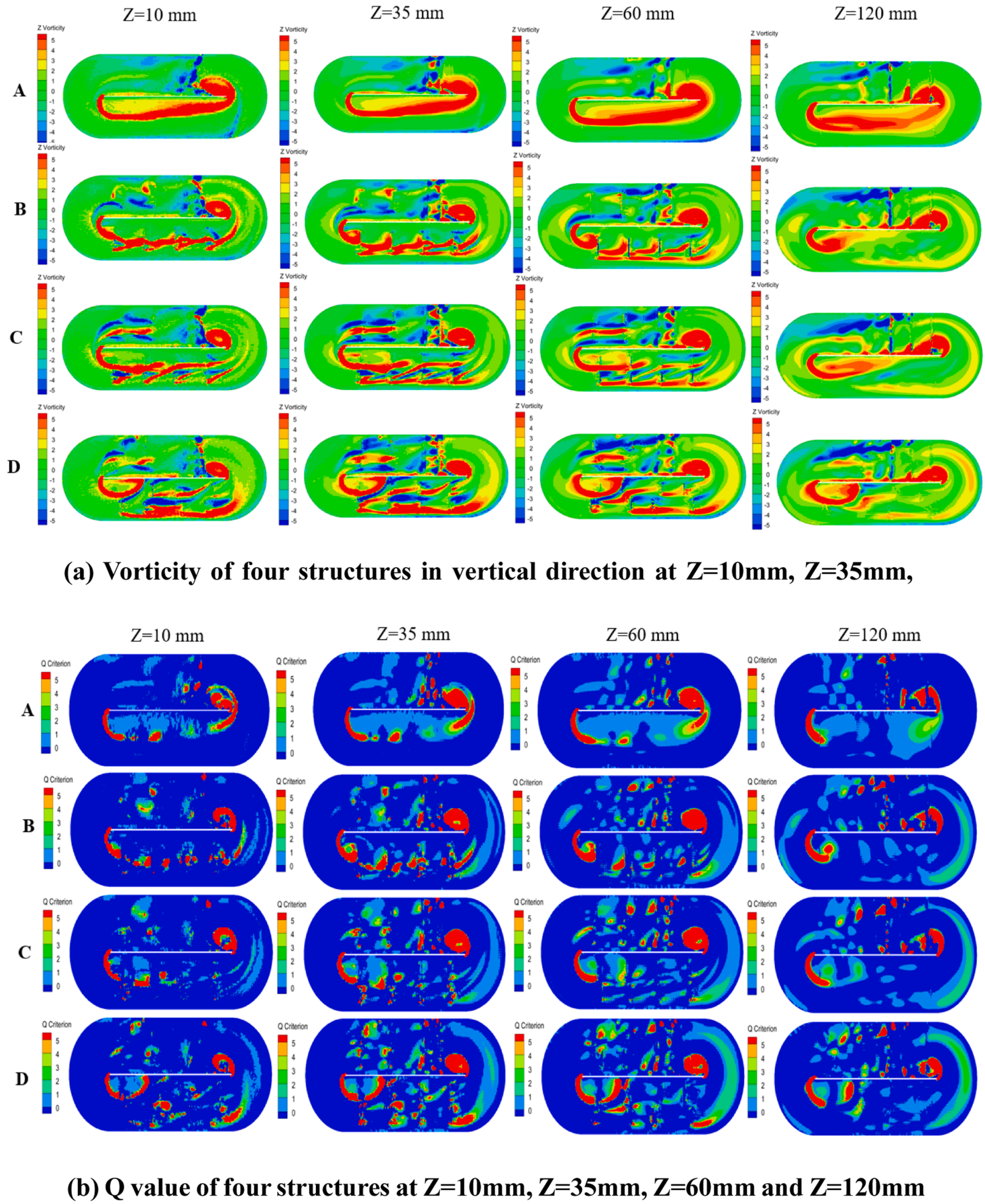


Fig. 4. (a) Vorticity of four structures in vertical direction at $Z = 10$ mm, $Z = 35$ mm, $Z = 60$ mm and $Z = 120$ mm. Fig 4(b) Q value of four structures at $Z = 10$ mm, $Z = 35$ mm, $Z = 60$ mm and $Z = 120$ mm.

Structure C, the fluid on the inner side of the baffle attracts the fluid on the outer side of the baffle resulting in a large area of low-velocity zones and deposition of algal cells in the outer side of the baffle. As for Structure A, although it brings similar results, it forms adverse pressure gradients to change the direction of the fluid so that the proportion of the low-velocity zone decreases.

3.3. RWP internal eddy analysis

The vortex flow is a basic structure of turbulence and described by

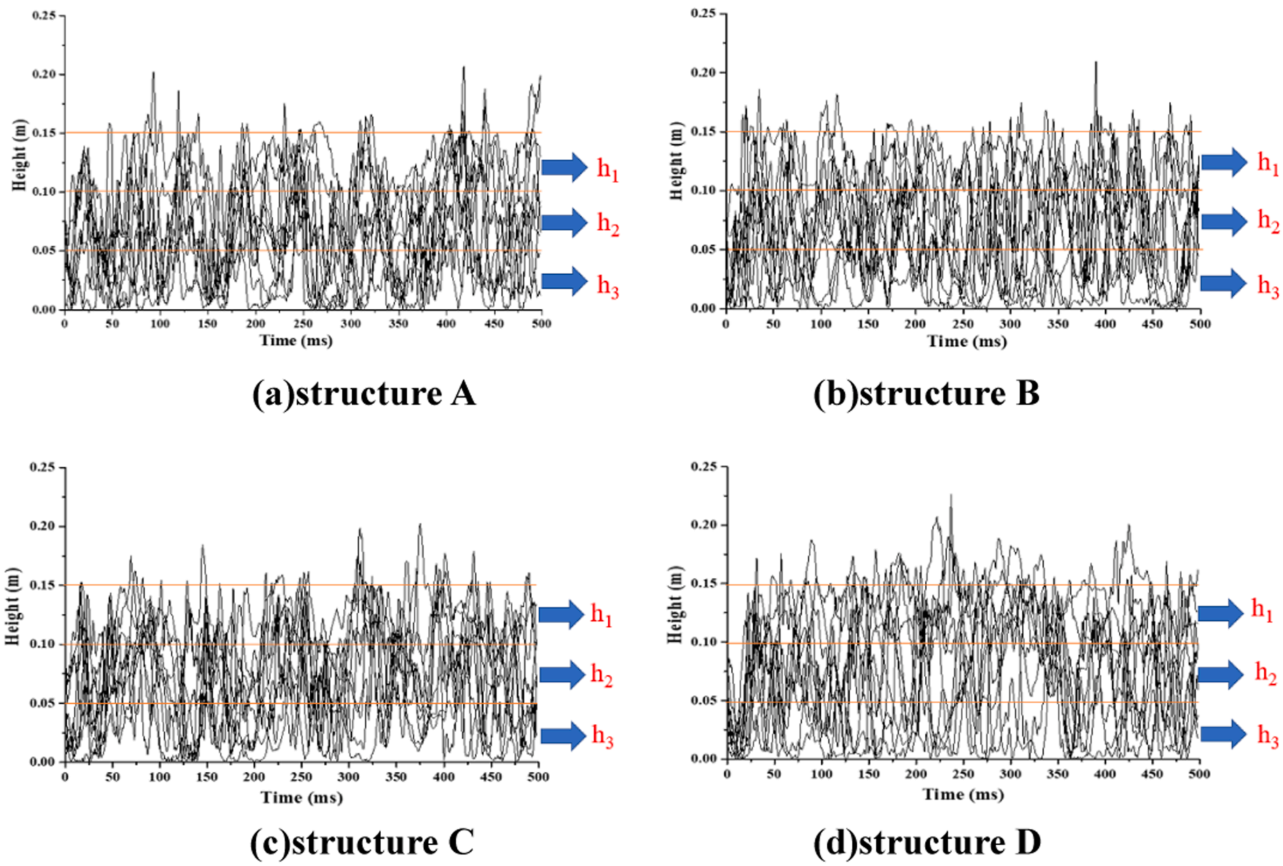


Fig. 5. Vertical change of single particle in RWP of four structures at 20 r/min (10 particles in each RWP are selected for clear visibility).

the vorticity field and the vortex. The vorticity field refers to the spatial distribution of vorticity, and the vortex is the circular current of the flow [20]. The motion of the vortex flow in the RWP is complicated and will influence the mixing of the fluid. Low intensity of the vortex will bring about the gathering of microalgae and barriers to the mixing of the fluid and mass transfer [21], and high intensity will come with high shear stress to the detriment of algal cells [22]. Fig. 4(a) depicts the vertical vorticity at different sections in the four types of RWPs. In the RWP without baffles, as the fluid flows through the curve to form the high-velocity zones in the outer side of the RWP and the low-velocity

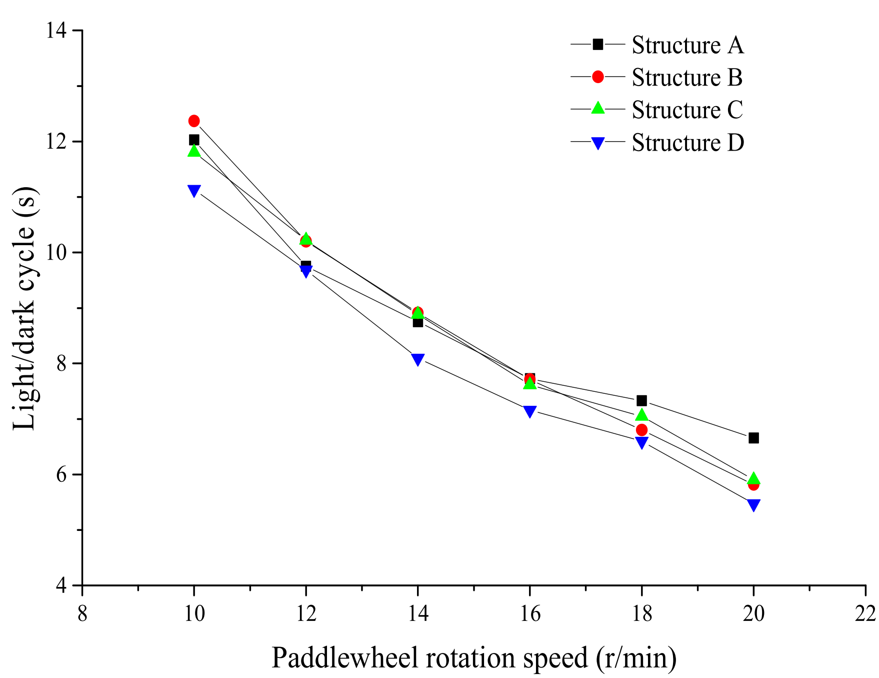


Fig. 6. Light dark cycle period of particles.

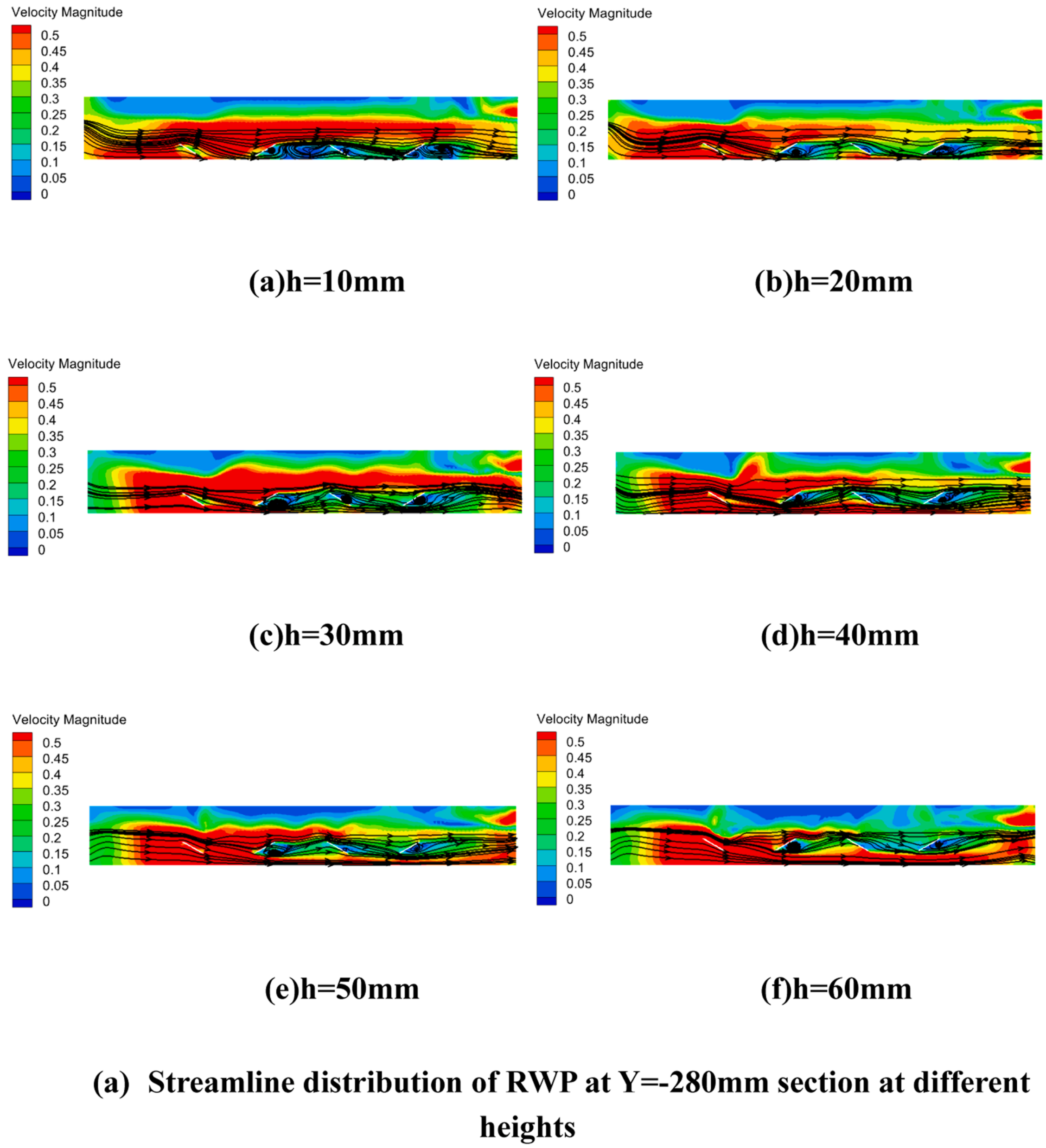


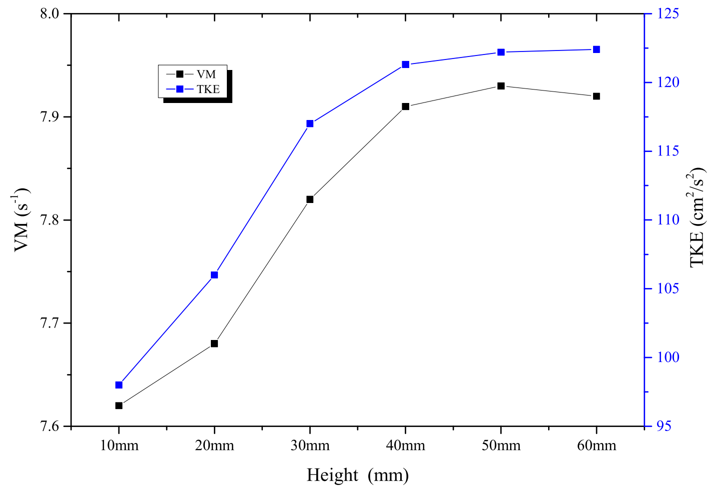
Fig. 7. (a) Streamline distribution of RWP at Y=−280 mm section at different heights. Fig 7(b) VM and TKE in RWP at different installation heights. Fig. 7(c) Mixing time and mass transfer coefficient of RWP at different heights.

zones near the baffles subject to the centrifugal force, the vortex flow of high intensity mainly appears in the turning area of both sides. The velocity difference between the two sides leads to high vorticity in the vertical direction in the middle of the direct channel. However, such high vorticity cannot facilitate the mixing of the fluid greatly. When it comes to Structure B and Structure C, they truly enable a more evenly distributed vorticity, but the high vorticity still mainly occurs in the outer side of the direct channel. Compared with the above mentioned structures, Structure D is able to change the direction of the flow constantly and create the even distribution of the vorticity on both sides to improve the overall mixing performance of the RWP.

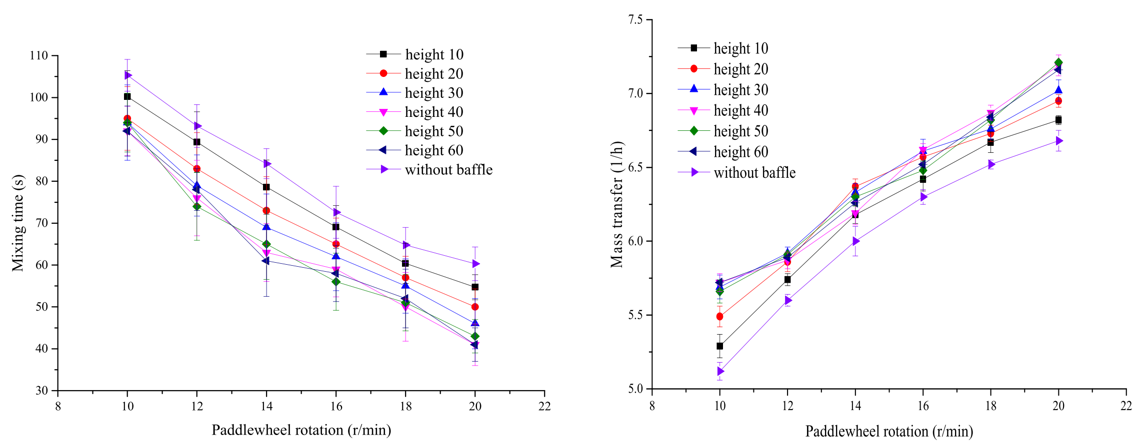
Fig. 4(b) shows the value of Q on different vertical sections in the four types of RWPs, and reflects the distribution of the vortex flow. In the RWP without baffles, the motion of the fluid is simple. The vortex flow mainly presents in the turning area of the curves on both sides and around the paddlewheels, less in the direct channels. When baffles are added to the RWP, the motion of the fluid is changed, and more vortex flows show up in the direct channels. All the three kinds of baffles proved able to create more even distribution of the vortex flow. However, compared with the RWPs with Structure B and Structure C, that with Structure D boasts a more favorable distribution of the vortex flow around the baffles in the middle, especially in the section with Z = 60 mm. In higher section, the influence of baffles on the fluid decreases. When the fluid flows away from the baffles, the baffles are no longer able to change the direction of the flow to form the vortex flow so that the vorticity is reduced gradually.

3.4. Particle motion analysis

Fig. 5 represents the motion of 10 random algal cell particles out of 300. To make it clearer, as shown in Fig. 2, the motion trajectory of particles is divided into three intervals: h1, h2, and h3. In the RWP



(b) VM and TKE in RWP at different installation heights



(c) Mixing time and mass transfer coefficient of RWP at different heights

Fig. 7. (*continued*).

without baffles, the particles crowd around the area with a vertical height below 0.1 m, and some of them leave the area at times. However, in the RWP with baffles, the area with a vertical height above 0.1 m sees a more even distribution of the particles in motion. Especially in the RWP of Structure D, more particles show up there.

The definition of the light-dark cycle of microalgae can be found in reference [23]. Fig. 6 depicts how the light-dark cycle of algal cells in the four RWPs of different structures changes with the speed of the paddlewheel with a light intensity of 60Klux and a concentration of the algal solution of 0.6 g/L. With the rise in the speed of the paddlewheel, the period of the light-dark cycle of the algal cells becomes shorter. When the speed of the paddlewheel increases from 10 r/min to 20 r/min, the frequency of the light-dark cycle in the RWP with cross baffles decreases from 11.1 s to 5.4 s. The light-dark cycle of the RWP with cross baffles is 18% lower than that without baffles, and 5.9% and 7.3% lower than that of Structure B and Structure C respectively.

3.5. Structural parameter optimization

3.5.1. Height

Fig. 7(a) displays the trajectory of the flow at Y=−280 mm (the screenshot of Y=−280 mm selected from the RWP) in the RWPs with baffles installed at different heights. When the baffle is installed at a lower position, that is, when the distance between the baffle and the bottom of the RWP is short, the fluid will encounter larger resistance when it reaches the baffles. It will then flow mainly from both sides of the baffle, which will form a large low-velocity zone in the backflow area of the baffle. It can be seen from Fig. 7 that a remarkable velocity retention zone is formed in the outer side of the baffle which is at $h = 10$ mm. With the increase in the height of the baffle, the proportion of the backflow area decreases. When the height of the baffle is above 40 mm, as shown in Fig. 7, there is no obvious difference in the size of the low-velocity zone. VM (Vorticity magnitude) is usually used to describe the rotation of the velocity vector. TKE (Turbulent Kinetic Energy) is the average kinetic energy per unit mass in turbulence, which is an important indicator for measuring the mixing effect of photobioreactors. Therefore, the larger the values of VM and TKE, the better the mixing of the flow field in the RWP and the more conducive to the growth of *Chlorella*. Besides, Fig. 7(b) is made to record the vorticity and the turbulence kinetic energy in the inner flow field. As it indicates, the vorticity and the turbulence kinetic energy rise with the height of the baffle, and do not change much when the baffle is installed at $h = 40$ mm or above due to limited positive influence of the baffle on the mixing of the fluid. Fig. 7(c) also records changes in the mixing time and the mass transfer coefficient. The increase in the rotation speed of the paddlewheel triggers reduction in the mixing time and rise in the mass transfer coefficient. When the speed of the baffle reaches 20 rpm and its installation height increases, the mixing time will decrease with a growing mass transfer coefficient. However, when the installation height of the baffle exceeds 40 mm, slight changes can be found in mixing time and

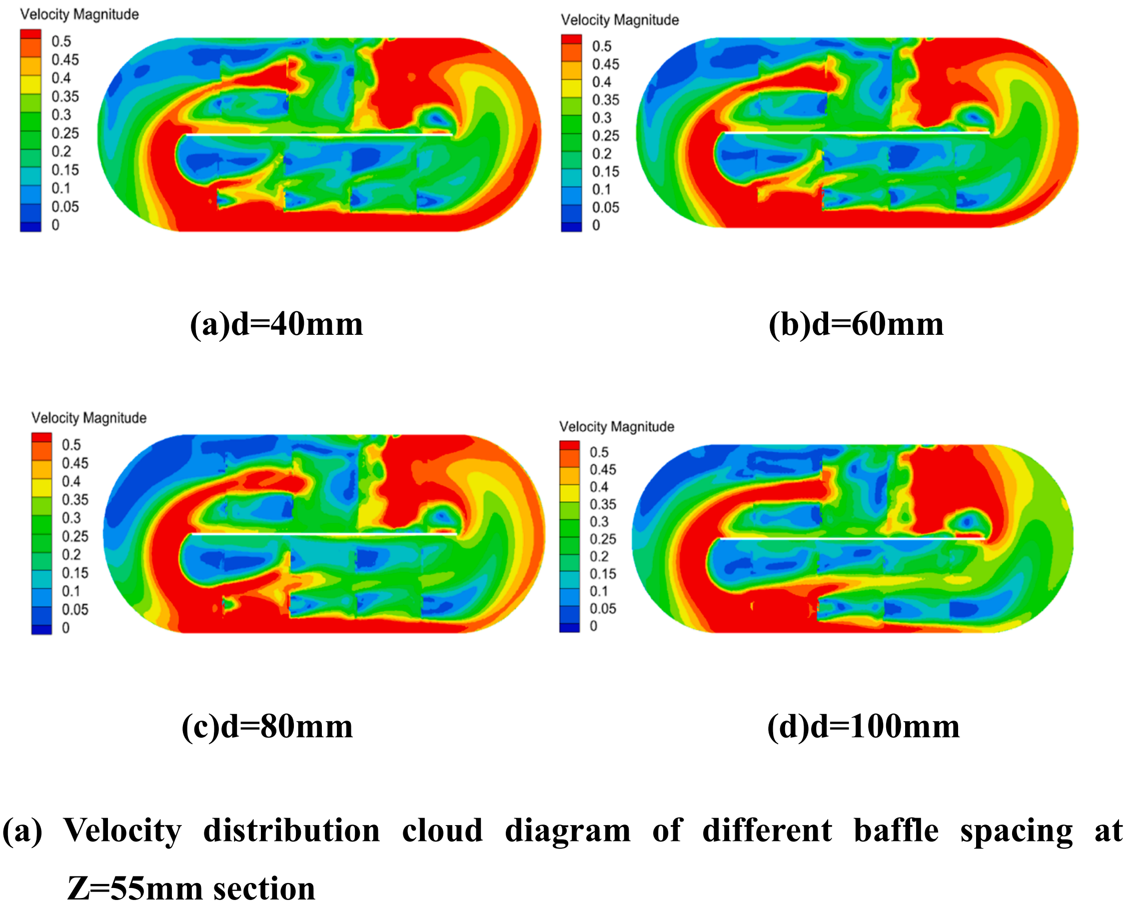


Fig. 8. (a) Velocity distribution cloud diagram of different baffle spacing at Z = 55 mm section 8(b) Vorticity and dead zone ratio in RWP with different baffle spacings 8(c) Mixing time and mass transfer coefficient of RWP at different spacings.

mass transfer coefficient.

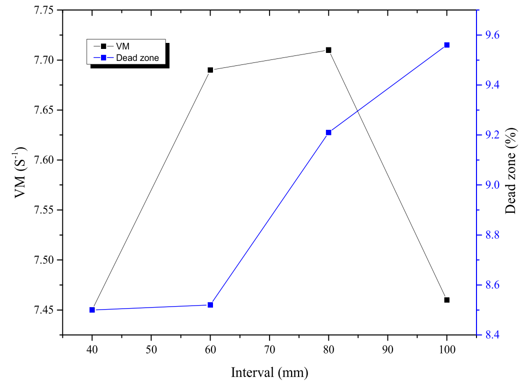
3.5.2. *Interval*

Fig. 8(a) presents the change in the flow velocity at $Z = 55$ mm with different distances (d) between baffles. As the distance between baffles increases, the ability of the baffle to change the direction of the fluid decreases. With large distance between baffles, the flow whose direction is changed by the baffles on the outer side has a lower velocity than the direct current in the middle of the RWP. Therefore, when the fluid that changes the flow direction passes through the middle, it is impacted by the fluid in direct-current direction, making it impossible for the fluid to complete the turning, causing poor mixing effect in the inner area of the direct-current channel. It can also be seen from Fig. 8(b) that the vorticity and the proportion of the dead zone vary with the distance between baffles. When the distance is enlarged, the vorticity increases first and then decreases, and the proportion of the dead zone becomes higher. The vorticity of the RWP with $d = 60$ mm is close to that of the RWP with $d = 80$ mm. However, when the distance changes from 60 mm to 80 mm, the proportion of the dead zone is increased by 8.1%, from 8.52% to 9.21%. In addition to above, Fig. 8(c) also registers changes in the mixing time and the mass transfer coefficient. As the rotation of the paddlewheel accelerates, the mixing time decreases and the mass transfer coefficient increases. When the rotation speed hits 20 r/min, the least mixing time of 43.7 s occurs at d = 60 mm, 16.3%, 7.8%, and 11.4% less than the mixing time at $d = 40$ mm, $d = 80$ mm, and $d = 120$ mm; and the mass transfer coefficient also reaches the maximum of 7.12% at d = 60 mm, 6.0%, 3%, and 7.7% greater than that at d = 40 mm, d = 80 mm, and d = 120 mm.

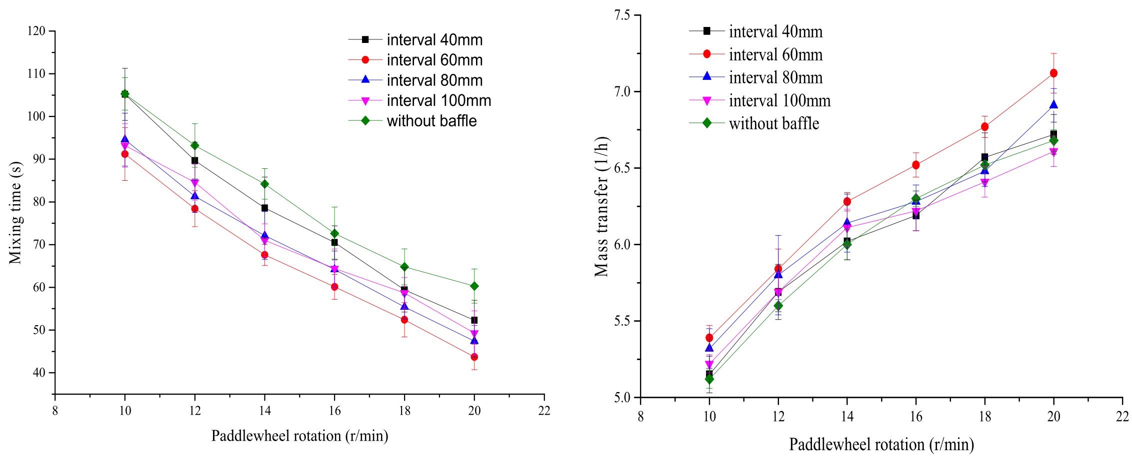
In order to verify the effect of structural optimization on the growth of *Chlorella vulgaris* in RWP, a seven day photoautotrophic contrast test of *Chlorella vulgaris* was conducted outdoors. The experimental results are shown in Fig. 9. The initial biomass of *Chlorella* is 0.4 g/L, and the rotating speed of paddle wheel is 20 r/min. The growth rate of *Chlorella* in RWP showed a trend of first increasing and then decreasing. On the seventh day of cultivation, the biomass of *Chlorella* in RWP with no baffle was 1.61 g/L, and that in RWP without optimized baffles was 1.98 g/L. After optimizing the baffle, the biomass of *Chlorella* increased to 2.2 g/L, by 36.6% and 11.1%, respectively. The results showed that the optimized cross baffles could increase the biomass of *Chlorella.*

3.5.3. *Conclusion*

The mixing of fluid in the RWP affects the mass transfer coefficient of *Chlorella* and the utilization of light energy by microalgae, and thus affects the yield of *Chlorella*. Based on these, an intersecting baffle plate was designed in this study to enhance the mixing ability of the RWP. According to the CFD simulation results, compared with the existing two types of co-direction baffle plate, the intersecting baffle plate can enlarge the distribution range of vortex, enhance its vorticity, and reduce the light-dark cycle of *Chlorella* in the RWP. Meanwhile, the installation position of the intersecting baffle plate was optimized, with its best experimental performance at a height of 40 mm and a spacing of 60 mm. Through this, the mixing time in the RWP was reduced by 31.7% and 27.5%, the mass transfer coefficient of *Chlorella* was enhanced by 7.6% and 6.6%, and the biological production of *Chlorella* was increased by 36.6% and 11.1%.



8(b) Vorticity and dead zone ratio in RWP with different baffle spacings



(c) Mixing time and mass transfer coefficient of RWP at different spacings

Fig. 8. (*continued*).

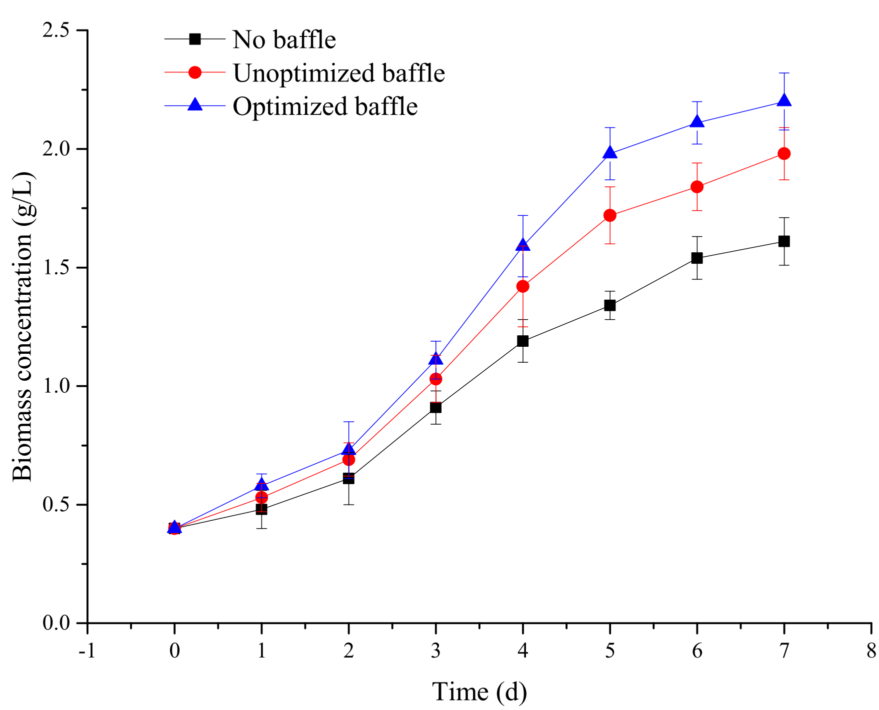


Fig. 9. Biomass of *Chlorella* in Outdoor Culture.

Author contribution

Shuangcheng Fu conceptualized the experiments and reviewed and edited the manuscript. Xiang Zhang designed the research, carried out the experiments, and wrote the original draft. Bin Dou and Chaolei Dai conducted the formal analysis and visualized data. Jie Ling conducted the experiments. Faqi Zhou and Yue Zhang provided methodology. Shenghu Yan revised the manuscript. All authors read and approved the manuscript.

Declaration of Competing Interest

The authors declare no competing interest.

Data availability

The data that has been used is confidential.

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